PHISHING EMAIL DETECTION & AWARENESS REPORT

Prepared For :

Future Interns

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Introduction

Phishing is a common cyberattack in which attackers pretend to be trusted individuals, companies, or organizations to trick people into sharing sensitive information such as usernames, passwords, bank details, or credit card information. These attacks are usually delivered through fake emails, text messages, websites, or phone calls that appear genuine and often create a sense of urgency to encourage quick action. The primary goal of phishing is to steal personal or financial information or to install malicious software on a victim's device. Understanding how phishing works and learning to identify suspicious messages are essential cybersecurity skills that help individuals and organizations protect themselves from online threats.

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Objectives

📧 Identify Phishing Emails

Learn to identify and differentiate phishing emails from legitimate emails by analyzing common phishing indicators and email security features.

🔍 Analyze Email Security

- Identify phishing emails by analyzing key email components.
- Evaluate sender details, domains, links, attachments, and message content for phishing indicators.
- Recognize common phishing techniques and social engineering tactics.
- Improve cybersecurity awareness and recommend preventive measures against phishing attacks.

⚠️ Understand Phishing Indicators

- Identify suspicious URLs, fake domains, urgent language, and credential requests.
- Detect malicious links, fake domains, urgency tactics, and requests for sensitive information.
- Recognize phishing indicators such as suspicious links, fake domains, urgency, and login requests.
- Spot fake URLs, spoofed domains, urgent messages, and credential theft attempts.

🛡️ Improve Cyber Awareness

- Develop the ability to recognize, identify, and avoid phishing attacks effectively.
- Increase cybersecurity awareness to identify and prevent phishing attempts.
- Enhance awareness of common phishing techniques and safe email practices.
- Build the knowledge and skills needed to detect and avoid phishing threats.

✅ Recommend

Recommend preventive strategies, including sender verification, link inspection, multi-factor authentication, and security awareness, to reduce phishing risks and protect sensitive information.

Methodology

Collect Email → Analyze Sender → Check Domain → Inspect Links → Identify Indicators → Assign Risk → Recommendation

Safe GitHub Security Notification

GitHub <noreply@github.com>
to me

Fri, 12 Jul 2026, 10:30 AM

GitHub

Hi Subhankar Mohanty,

We detected a new sign-in to your GitHub account.

Date: 12 July 2026, 10:30 AM IST
Location: Bhubaneswar, India
Device: Windows 11 / Chrome

If this was you, you can safely ignore this email.
If this wasn't you, please secure your account.

Review Activity

Thanks,
The GitHub Security Team

Educational Example – Fictional email created for cybersecurity awareness and educational purposes.

Analysis – Email 1 (Safe)

Criteria	Observation
Sender Email	noreply@github.com
Domain Authenticity	Official GitHub Domain
Language/Tone	Professional
Links/Buttons	Official GitHub Link
Request for Sensitive Information	No
Urgency	No

Tools Used:

- Gmail
- Google
- Chrome
- Canva
- GitHub

GitHub Security Notification (SAFE)

Risk Level

🟢 Low

Reason:

This email is sent from the official GitHub domain and does not contain any suspicious links or requests for sensitive information.

Classification

✅ Safe Email

Recommendations:

- Verify the sender if unsure.
- Keep MFA enabled.
- Use the official GitHub website.

Microsoft Password Reset Email Screenshot

Password reset request for your Microsoft account Inbox ×

Microsoft Support <support.microsoft.account@gmail.com> Fri, 12 Jul 2026, 01:15 PM

to me

Microsoft

Hello,

We received a request to reset the password for your Microsoft account.

If you did not make this request, you can cancel it now.

Cancel Password Reset

If the button above does not work, copy and paste this link in your browser:

http://security-microsoft-account.com/reset

Thanks,

Microsoft Account Team

(Microsoft Password Reset Email (Educational Example))

Educational Disclaimer:

This email is a fictional example created solely for cybersecurity awareness and educational purposes as part of the Future Interns Cyber Security Internship. It does not represent a real Microsoft email or phishing campaign.

Analysis – Microsoft Password Reset Email (Suspicious)

Criteria	Observation
Sender Email	support.microsoft.account@gmail.com (Free Gmail Address ❌)
Domain Authenticity	Uses **gmail.com** instead of an official Microsoft domain ❌
Language / Tone	Professional but unexpected password reset request ⚠️
Links / Buttons	Contains a password reset button; verify before clicking ⚠️
Request for Sensitive Information	May redirect users to enter account credentials ⚠️
Urgency / Threat	Medium – encourages immediate action ⚠️

Tools Used:

- Gmail
- Google
- Chrome
- Canva
- GitHub

Risk Level

🟡 Medium

Reason:

The email appears professional but uses a free Gmail address rather than an official Microsoft domain. Users should verify password reset requests directly through the official Microsoft website instead of clicking email links.

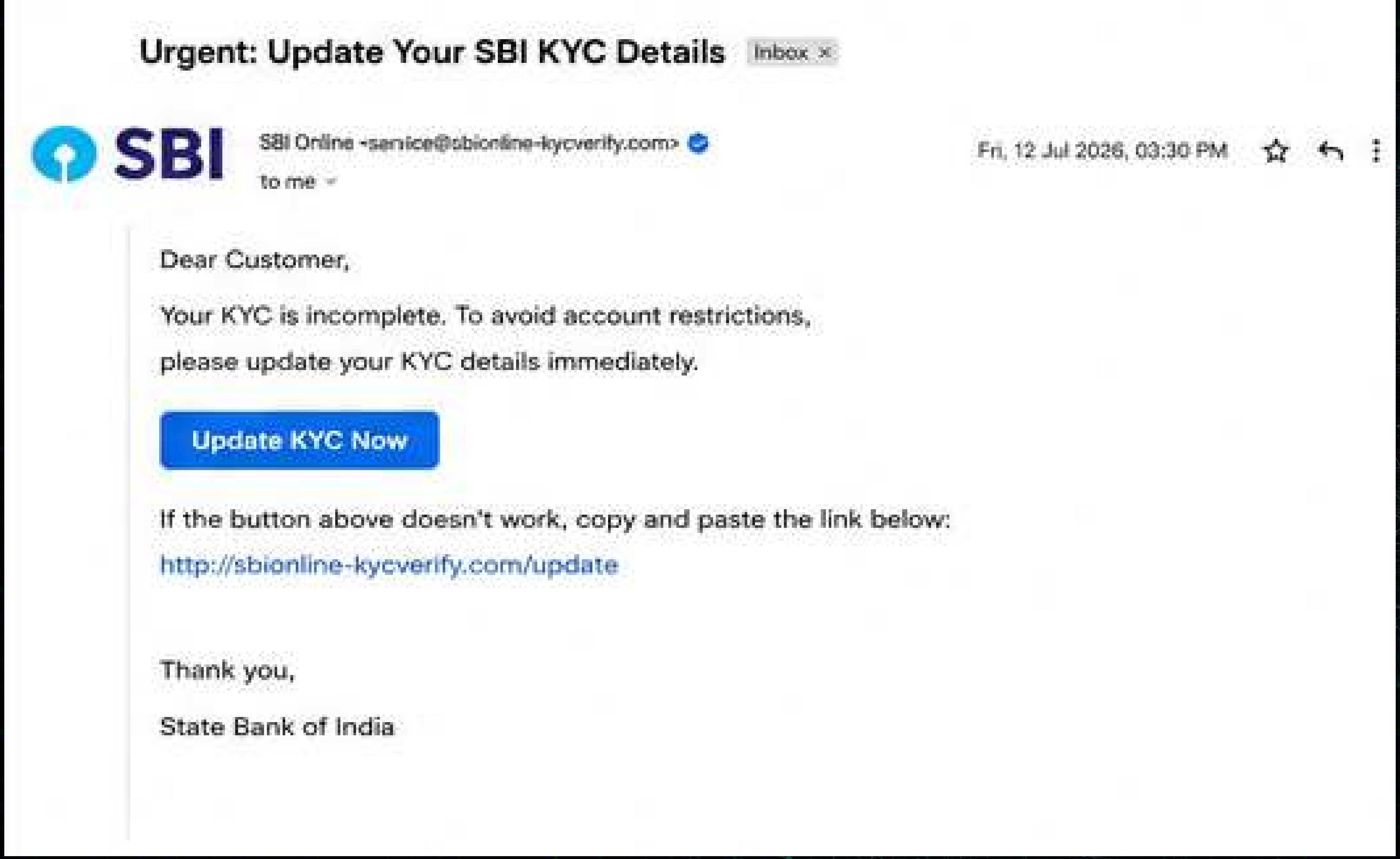
Classification

🟡 Suspicious Email

Recommendation:

- Verify the sender's email address carefully.
- Avoid clicking password reset links without verification.
- Visit the official Microsoft website manually.
- Enable Multi-Factor Authentication (MFA).
- Report unexpected password reset emails.

SBI KYC VERIFICATION EMAIL SCREENSHOT



Urgent: Update Your SBI KYC Details Inbox ×

SBI

SBI Online <service@sbionline-kycverify.com>

to me

Fri, 12 Jul 2026, 03:30 PM

Dear Customer,

Your KYC is incomplete. To avoid account restrictions, please update your KYC details immediately.

Update KYC Now

If the button above doesn't work, copy and paste the link below:

http://sbionline-kycverify.com/update

Thank you,

State Bank of India

(SBI KYC VERIFICATION EMAIL (EDUCATIONAL EXAMPLE))

EDUCATIONAL DISCLAIMER:

THIS EMAIL IS A FICTIONAL EXAMPLE CREATED SOLELY FOR CYBERSECURITY AWARENESS AND EDUCATIONAL PURPOSES AS PART OF THE FUTURE INTERNS CYBER SECURITY INTERNSHIP. IT DOES NOT REPRESENT A REAL SBI COMMUNICATION OR PHISHING CAMPAIGN.

ANALYSIS – SBI (PHISHING)

Criteria	Observation
Sender Email	service@sbionline-kycverify.com (Fake Domain ❌)
Domain Authenticity	Not an official SBI domain ❌
Language / Tone	Urgent and threatening ❌
Links / Buttons	Suspicious "Update KYC Now" button ❌
Request for Sensitive Information	Requests banking/KYC information ❌
Urgency / Threat	High – claims account restrictions ❌

Tools Used:

- Gmail
- Google
- Chrome
- Canva
- GitHub

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Risk Level

🔴 High

Reason:

The email uses a fake banking domain and creates urgency to trick users into revealing sensitive banking information through a fraudulent website.

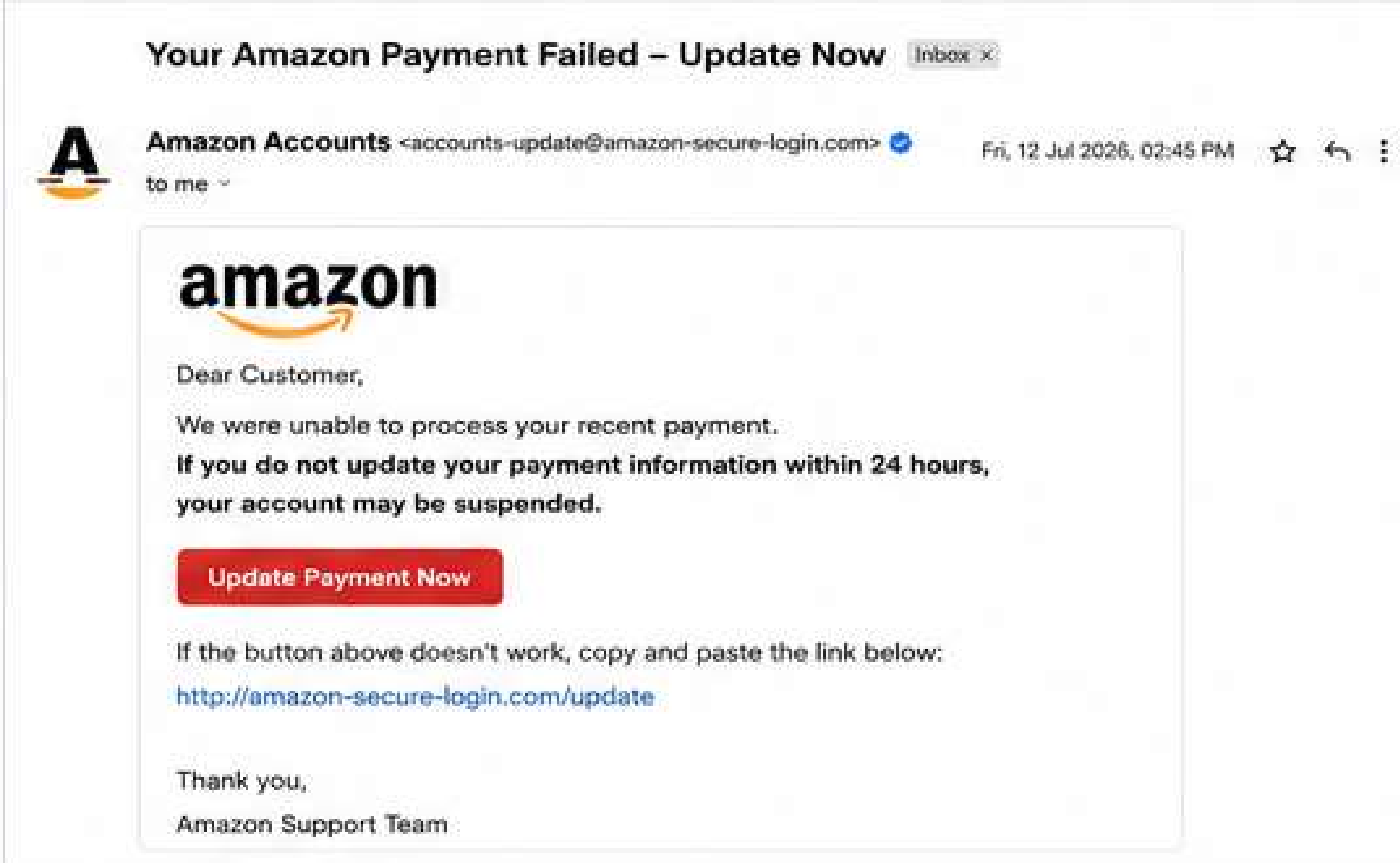
Classification

🔴 Phishing Email

Recommendation:

- Never click the "Update KYC Now" button.
- Visit the official SBI website or mobile banking app directly.
- Never share banking credentials or OTPs through email.
- Report the email to the bank's security team.
- Delete the email after reporting it.

Amazon Payment Failed Phishing Email screenshort



Educational Disclaimer:

This email is a fictional example created solely for cybersecurity awareness and educational purposes as part of the Future Interns Cyber Security Internship. It is designed to demonstrate common phishing techniques and does not represent an actual email from Amazon.

Phishing Amazon Payment Failed Email (Educational Example)

Analysis –Amazon Payment Failed

Criteria	Observation
Sender Email	accounts-update@amazon-secure-login.com (Fake Domain ❌)
Domain Authenticity	Uses a fake domain instead of the official Amazon domain ❌
Language / Tone	Urgent and threatening; pressures the user to act immediately ❌
Links / Buttons	Contains a suspicious "Update Payment Now" button ❌
Request for Sensitive Information	Requests payment or account credentials ❌
Urgency / Threat	High – warns that the account may be suspended ❌

Tools Used:

- Gmail
- Google
- Chrome
- Canva
- GitHub

Risk Level

🔴 High

Reason:

The email uses a fake sender domain and creates urgency by claiming that the user's payment has failed and the account may be suspended. It contains a suspicious button that may redirect users to a fraudulent website to steal login credentials or payment information.

Classification

🔴 Phishing Email

Recommendation:

- Do not click the "Update Payment Now" button.
- Access Amazon only through the official website or mobile application.
- Never enter payment details through links received in unsolicited emails.
- Verify your account status directly from your Amazon account.
- Report the email as phishing and delete it immediately.

Comparison of Email Analysis

Criteria	Email 1 (GitHub)	Email 2 (Microsoft)	Email 3 (Amazon)	Email 4 (SBI)
Classification	✅ Safe	⚠️ Suspicious	❌ Phishing	❌ Phishing
Risk Level	🟢 Low	🟡 Medium	🔴 High	🔴 High
Sender Domain	Official (github.com)	Free Gmail Address	Fake Amazon Domain	Fake SBI Domain
Domain Authenticity	Legitimate	Suspicious	Fake	Fake
Urgency	No	Medium	High	High
Suspicious Links	No	Possible	Yes	Yes
Requests Sensitive Information	No	Possible	Yes	Yes
Recommended Action	Verify if unsure	Verify via official website	Do not click; report	Do not click; report

Key Phishing Indicators

Phishing emails often contain warning signs that help identify malicious intent. The following indicators were observed during the analysis of the sample emails.

Indicator	Description
📧 **Fake Sender Email**	The sender's email address does not belong to the official organization.
🌐 **Suspicious Domain**	The domain name is misspelled or different from the legitimate website.
⚠️ **Urgent Language**	The email creates fear or pressure to force immediate action.
🔗 **Suspicious Links**	Hyperlinks redirect to fake or untrusted websites.
🔒 **Request for Sensitive Information**	Asks for passwords, OTPs, banking details, or personal information.
📎 **Unexpected Attachments**	Includes files that may contain malware or malicious code.
❗ **Grammar or Spelling Errors**	Poor grammar, spelling mistakes, or unusual formatting.
🎭 **Impersonation**	Pretends to be from a trusted company or organization to gain trust.

Prevention Tips

Best Practices:

✅ Always verify the sender's email address before responding.
🌐 Check the domain name carefully for spelling mistakes or unusual extensions.
🔗 Hover over links before clicking to verify the destination.
🔒 Never share passwords, OTPs, banking details, or personal information through email.
🛡️ Enable Multi-Factor Authentication (MFA) for important accounts.
💻 Keep your operating system, browser, and antivirus software updated.
🚫 Avoid downloading unexpected attachments from unknown senders.
📢 Report suspicious emails to your organization's IT or security team.
🌍 Access websites by typing the official URL instead of clicking email links.
📚 Stay informed about common phishing techniques through regular cybersecurity awareness training.

Final Security Message:

Think Before You Click.
Always verify the sender, inspect the domain, and avoid sharing sensitive information. A few seconds of verification can prevent credential theft, financial fraud, and data compromise.

Key Findings:

50% of the analyzed emails were identified as phishing attempts, using fake domains, urgent language, and credential theft tactics.
Legitimate emails consistently used trusted sender domains, secure HTTPS links, and did not request sensitive information.
Suspicious emails displayed common phishing indicators, including spoofed email addresses, generic greetings, misleading hyperlinks, and requests for immediate action.
Verifying the sender's email address, domain name, embedded links, and attachments proved to be the most effective method for distinguishing phishing emails from legitimate ones.
User awareness and careful email verification are essential to prevent phishing attacks, protect sensitive information, and reduce the risk of credential compromise.

Conclusion:

This project successfully demonstrated how phishing emails can be identified by analyzing key indicators such as the sender's email address, domain authenticity, message content, urgency, embedded links, and requests for sensitive information. Through the examination of four educational email examples, it became clear that phishing attacks often rely on social engineering techniques to deceive users into revealing confidential information.

The analysis highlighted the differences between safe, suspicious, and phishing emails, emphasizing the importance of carefully verifying email authenticity before taking any action. By following cybersecurity best practices, such as checking sender details, avoiding suspicious links, enabling Multi-Factor Authentication (MFA), and reporting suspicious emails, users can significantly reduce the risk of becoming victims of phishing attacks.

Overall, this task enhanced practical knowledge of phishing detection and reinforced the importance of cybersecurity awareness in protecting personal and organizational information from email-based threats.

References:

- GitHub Security Documentation – https://docs.github.com/en/code-security
- Microsoft Security – https://www.microsoft.com/security
- Amazon Customer Security & Privacy – https://www.amazon.com/gp/help/customer/display.html?nodeId=GKM69DUUYKQWKWX7
- SBI Cyber Security Awareness – https://sbi.co.in/web/customer-care/security-tips
- OWASP – Phishing Guidance – https://owasp.org
- CISA – Phishing Guidance – https://www.cisa.gov/topics/cybersecurity-best-practices/phishing-guidance
- Google Safety Center – https://safety.google
- National Institute of Standards and Technology (NIST) Cybersecurity Framework – https://www.nist.gov/cyberframework

THANK YOU

GET IN TOUCH

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